

2520090016_P.Rahul

OSSP:

Commands:

Uname : It displays the System information

Ls : It displays the complete CPU information

```
root@Rahul:~# nano process.c
root@Rahul:~# gcc process.c -o process
root@Rahul:~# ./process
Enter Linux command: ls

Child Process
Child PID = 525
File2.txt      copyfile.c file.c.save process.c      s4task2
s4task4      skill4task3.c task4 week3.c
OSSP          cpro      file.txt process.c.save s4task2.c
s4task4.c    task      task4.c
command_executor cpro.c filename s4task      s4task3
s4task5      task1.c week1
command_executor.c file filename.c s4task1
s4task3. s4task5.c task3 week1.c
copyfile      file.c process s4task1.c s4task3.c
skill4task3 task3.c week3

Parent Process
Parent PID = 524
Child PID = 525
root@Rahul:~#
```

Ps : It displays the running process of our file

```
root@Rahul:~# nano process.c
root@Rahul:~# gcc process.c -o process
root@Rahul:~# ./process
Enter Linux command: ps

Child Process
Child PID = 550
  PID TTY          TIME CMD
  499 pts/0    00:00:00 bash
  549 pts/0    00:00:00 process
  550 pts/0    00:00:00 ps

Parent Process
Parent PID = 549
Child PID = 550
root@Rahul:~#
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among multiple programs by giving each one a turn to execute.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It processes both Chrome and music player.

2. Memory (RAM) -> Memory Management

The operating system gives memory to each program that is running
It makes sure one program does not use another program's memory.
Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system stores and organizes files on the storage device.
Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.